

YANN HALLOUARD

Senior Data Scientist & ML Engineer

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EXPERIENCE

Senior Data Scientist

TotalEnergies Digital Factory

Apr 2025 – Present

Paris

DataTeam - InnovationLab: Co-produced AI products with Mistral

- LLM production pipelines: Robust generative AI industrialization
- Parallel implementation: Vocal chat service for end users & report generator for chemical refinery
- Technology stack: Temporal.io, Kubernetes, Entity Matching, Production Pipelines

Senior Data Scientist

TotalEnergies Digital Factory

Feb 2023 – Apr 2025

Paris

SAFEWORLD - Real-time Industrial Safety System

- AI-powered safety compliance monitoring for 10 industrial sites
- Computer Vision optimization: RT-DETR, DinoV2 for real-time inference
- Edge computing architecture: 8-second event-to-platform detection latency
- Model performance: mAP@0.5 of 0.8 (35m), 0.6 (70m)
- Technology stack: DynamoDB, AWS Lambda, Serverless Architecture

Data Scientist

TotalEnergies Digital Factory

Nov 2021 – Jan 2023

Paris

DEGAS - Topographic Signal Denoising Platform

- ML denoising system for landslide risk monitoring sensors
- Change Point Detection: 96% accuracy on multivariate time series
- Autoencoder-based correction with continuous deployment
- Technology stack: Azure ML, TimescaleDB, Azure Functions

Data Scientist

TotalEnergies Digital Factory

Apr 2021 – Nov 2021

Paris

RADAR - Oil Well Production Anomaly Detection

- Bayesian anomaly detection for production forecasting enhancement
- Operational noise filtering from genuine production issues
- Technology stack: Databricks, Azure Data Factory, MLflow, PostgreSQL

MY LIFE PHILOSOPHY

"Graduated from École des Mines de Saint-Étienne, passionate about transforming models into tangible and usable solutions. With solid experience in production deployment, motivated to continue developing products that bring real added value to users."

MOST PROUD OF



TDF GenAI Hackathon

2nd Place - Oct 2024



Azure Certification

DP200 - Sept 2022



Assistant Secretary CSE

TotalEnergies Digital Factory - Mar 2022 to present - Managing budget and employee programs for 150+ Digital Factory employees



Community Contributions

Sktime, Moto, Roboflow

MENTORING

- **Team & intern supervision:** Data technical leadership on delivered products and intern mentoring across multiple projects
- **Technical community animation:** Organizing community of practices within Digital Factory
- **Recruitment & interviews:** Conducted 30+ technical interviews for Data Science and ML Engineering positions
- **Training & seminars:** Delivered technical presentations to audiences up to 300 people

SKILLS



Junior Data Scientist

TotalEnergies Digital Factory

📅 Jun 2020 – Apr 2021

📍 Paris

- Energy efficiency analysis via electricity consumption clustering
- FPSO CO2 optimization using causal analysis and multivariate classification
- Helicopter crew rotation optimization (VRP with time windows)
- Technology stack: Python, Streamlit, OR-Tools, Statistical Modeling

PROJECTS

Héméa - Mobile Health App

Personal Project

📅 May 2025 – Present

- Mobile application for organizing, storing and visualizing biological analyses
- User-friendly interface for medical test results management
- Data visualization dashboard for health monitoring trends
- Secure storage and privacy-focused architecture
- Technology stack: Mistral, React Native, Expo Go, SQLite

PUBLICATIONS

📄 Articles

- Y. Hallouard, "Migrate your poetry project to uv," *TotalEnergies Digital Factory - Medium*, 2025, 1.6K views.
- Y. Hallouard, "Cultural mindset: A thought about collective ownership," *TotalEnergies Digital Factory - Medium*, 2024, 0.5K views.
- Y. Hallouard, "Open ai: Multi images prompting, python api," *TotalEnergies Digital Factory - Medium*, 2024, 1.2K views.
- Y. Hallouard, "Zoedepth: 3d visualization of depth estimation," *TotalEnergies Digital Factory - Medium*, 2024, 1.K views.
- Y. Hallouard, "Sktime — timeseriesforestclassifier improving features comprehension," *TotalEnergies Digital Factory - Medium*, 2023, 1.1K views.
- Y. Hallouard, "Sktime — feature importance on timeseriesforestclassifier," *TotalEnergies Digital Factory - Medium*, 2022, 4.5K views.
- Y. Hallouard, "Create your own "google" with sphinx," *Medium*, 2021, 0.4K views.
- Y. Hallouard, "End-to-end anomalies detection models evaluation algorithms," *TotalEnergies Digital Factory - Medium*, 2020, 1.1K views.
- Y. Hallouard, "Time series: The problem with resampling," *TotalEnergies Digital Factory - Medium*, 2020, 4.3K views.

LANGUAGES

Français



English - TOEIC 825



EDUCATION

Engineering Degree

École Nationale Supérieure des Mines de Saint-Étienne

📅 Completed in 2019

Data Science, Big Data, ROAD specialization
Gestion et Finance

Master of Business Intelligence and Innovation Management

IAE Saint-Étienne

📅 Completed in 2019

ML in Embedded Systems

Seoul National University

📅 Completed in 2019

Classe Préparatoire - PCSI/PSI

Lycée Bellevue

📅 Completed in 2017

REFEREES

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